EE4-08
EE9-SO14

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2012

MSc and EEE/ISE PART IV: MEng and ACGI

DIGITAL IMAGE PROCESSING

Friday, 10 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer THREE questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	P.T. Stathaki
	Second Marker(s) :	C. Mehring

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1. a) i) Explain when a two dimensional image transform is symmetric, separable and unitary. **[2]**

ii) Discuss the property of energy compaction of a transform. **[2]**

b) Let $f(x,y)$ denote an $M \times N$-point 2-D sequence that is zero outside $0 \le x \le M-1$, $0 \le y \le N-1$, where M and N are integer powers of 2. In implementing the standard Walsh Transform of $f(x,y)$, we relate $f(x,y)$ to a new $M \times N$-point sequence $W(u,v)$.

i) Define the sequence $W(u,v)$ in terms of $f(x,y)$. **[2]**

ii) Define the concept of sequency in the Walsh transform. **[2]**

iii) Discuss three advantages of the two dimensional Walsh transform. **[2]**

c) Consider the population of random vectors $\underline{f}$ of the form

$$\underline{f} = \begin{bmatrix} f_1(x,y) \\ f_2(x,y) \\ \vdots \\ f_n(x,y) \end{bmatrix}.$$

Each component $f_i(x,y)$ represents an image. The population arises from their formation across the entire collection of pixels. Suppose that $n \ge 2$ and n even. Consider now a population of random vectors of the form

$$\underline{g} = \begin{bmatrix} g_1(x,y) \\ g_2(x,y) \\ \vdots \\ g_n(x,y) \end{bmatrix}$$

where the vectors $\underline{g}$ are the Karhunen-Loeve (KL) transforms of the vectors $\underline{f}$.

i) Derive the relationship between $\underline{g}$ and $\underline{f}$. Explain clearly how we formulate the transformation matrix of the KL transform. **[2]**

ii) Discuss the form of the covariance matrix of $\underline{g}$ and the physical meaning of its elements. **[2]**

iii) Suppose that some elements of the covariance matrix of $\underline{g}$ are very small. Discuss the effectiveness of the KL transform in that case. **[2]**

iv) Suppose that the elements of the covariance matrix of $\underline{g}$ are very similar. Discuss the effectiveness of the KL transform in that case. **[2]**

v) Suppose that a credible job could be done of reconstructing approximations to the n original images by using only the half of the principal component images associated with the largest eigenvalues. What would be the mean square error incurred in doing so? Express your answer as a percentage of both the maximum and the minimum possible error. **[2]**

2. a) A 4×4 grey scale original image passes through three spatial shift invariant linear filters, resulting in three filtered images shown in **Figure 2.1** below.

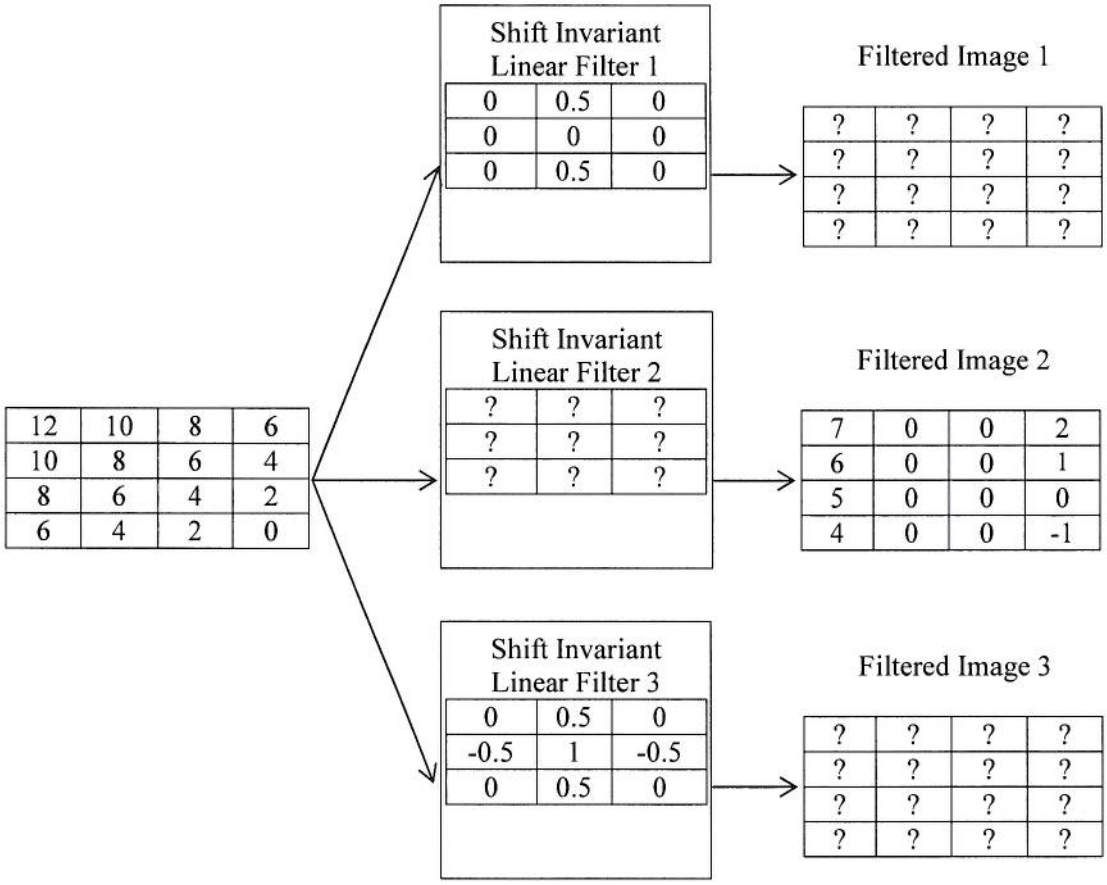


Figure 2.1

i) Compute Filtered Image 1 using zero-padding of the original image. **[2]**
ii) Compute Filtered Image 3 using zero-padding of the original image. **[2]**
iii) Based on relationship between Filtered Image 1, Filtered Image 2 and Filtered Image 3, determine the filter coefficients in Shift Invariant Linear Filter 2. **[2]**

b) Provide linear filter masks for the following operations:
i) Region averaging. **[2]**
ii) Weighted region averaging (with pixels near the centre given more weight). **[3]**
iii) Detecting vertical (horizontal) edges. **[2]**
iv) Remove salt and pepper noise. **[2]**
v) Image sharpening. **[2]**
vi) Image sharpening while preserving a proportion of the original image. **[3]**

3. We are given the degraded version g of an image f such that in lexicographic ordering

$$g = Hf + n$$

where H is the degradation matrix which is assumed to be block-circulant, and n is the noise term which is assumed to be zero mean, independent and white. We propose to restore the degraded image using Wiener filtering.

a) i) Derive the expressions for both the Wiener filter estimator and the restored image in both spatial and frequency domains and explain all symbols used. **[5]**

ii) The autocorrelation matrix of the original image f and of the additive noise n are required for Wiener filtering. Since both f and n are unknown, suggest a way to obtain estimates of them. Justify your answer. **[5]**

b) Describe the Wiener smoothing filter. **[5]**

c) Describe the circumstances in which the Wiener filter is related to the Inverse Filter. **[5]**

4. a) Assume three grey scale images A, B, C which have 8 different grey levels denoted by $R_i, i = 0,\ldots,7$. The histograms of the three images are given in the **Table 4.1** below.

Grey Level	Probability Image A	Probability Image B	Probability Image C
R_0	0.15	0.32	0.125
R_1	0.11	0.16	0.125
R_2	0.19	0.09	0.125
R_3	0.15	0.05	0.125
R_4	0.08	0.14	0.125
R_5	0.02	0.15	0.125
R_6	0.2	0.01	0.125
R_7	0.1	0.08	0.125

Table 4.1

For each of the above images A, B, C, derive the Huffman code and answer the following questions:

i) What's the entropy of the original data source? **[2]**

ii) What's the average coding length if using fixed-length coding? **[2]**

iii) What's the average coding length of Huffman coding? (Need to show details of Huffman coding.) **[4]**

iv) Comment on relationship between probability combination, entropy and redundancy. **[4]**

b) The following **Figure 4.1** shows an 8×8 image with 4 different grey levels (black, dark grey, light grey, white).

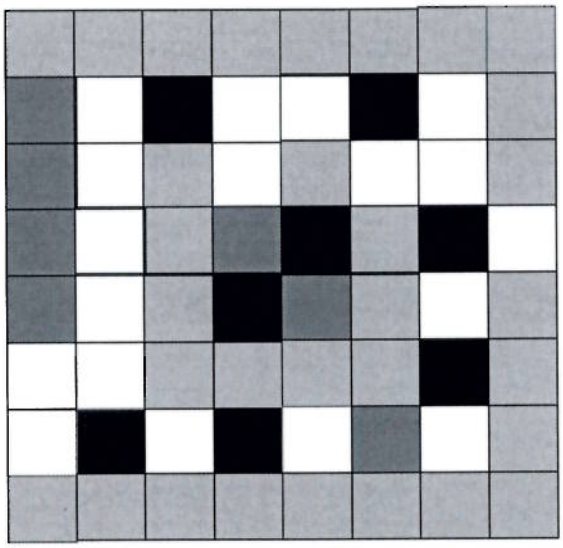


Figure 4.1

i) Derive the probability of appearance (that forms the histogram) for each intensity (grey) level. Calculate the entropy of this image. **[2]**

ii) Derive the Huffman code. **[2]**

iii) Calculate the average length of the fixed length code and that of the derived Huffman code. **[2]**

iv) Calculate the ratio of image size (in bits) between using the fixed length coding and Huffman coding. Calculate the relative coding redundancy. **[2]**

Solutions 2012

EE4 - 08
EE9 - 2014

1. a) i) Explain what is unitary transform.

For an $N \times N$ image $f(x,y)$ the forward and inverse transforms are given below

$$g(u,v) = \sum_{x=0}^{N-1}\sum_{y=0}^{N-1} T(u,v,x,y) f(x,y)$$

$$f(x,y) = \sum_{u=0}^{N-1}\sum_{v=0}^{N-1} I(x,y,u,v) g(u,v)$$

where, again, $T(u,v,x,y)$ and $I(x,y,u,v)$ are called the forward and inverse transformation kernels, respectively. The forward kernel is said to be **separable** if

$$T(u,v,x,y) = T_1(u,x) T_2(v,y)$$

It is said to be **symmetric** if T_1 is functionally equal to T_2 such that

$$T(u,v,x,y) = T_1(u,x) T_1(v,y)$$

The same comments are valid for the inverse kernel. If the kernel $T(u,v,x,y)$ of an image transform is separable and symmetric, then the transform $g(u,v) = \sum_{x=0}^{N-1}\sum_{y=0}^{N-1} T(u,v,x,y) f(x,y) = \sum_{x=0}^{N-1}\sum_{y=0}^{N-1} T_1(u,x) T_1(v,y) f(x,y)$ can be written in matrix form as follows

$$\underline{g} = \underline{T}_1 \cdot \underline{f} \cdot \underline{T}_1^T$$

where $\underline{f}$ is the original image of size $N \times N$, and $\underline{T}_1$ is an $N \times N$ transformation matrix with elements $t_{ij} = T_1(i,j)$. If, in addition, $\underline{T}_1$ is a unitary matrix, i.e., $\underline{T}^{-1} = \underline{T}^{*T}$, then the transform is called **separable unitary** and the original image is recovered through the relationship

$$\underline{f} = \underline{T}_1^{*T} \cdot \underline{g} \cdot \underline{T}_1^{*}$$

ii) Discuss the property of energy compaction of a transform.

Most unitary transforms pack a large fraction of the energy of the image into relatively few of the transform coefficients. This means that relatively few of the transform coefficients have significant values and these are the coefficients that are close to the origin (small index coefficients). This property is very useful for compression purposes.

c) Let $f(x,y)$ denote an $M \times N$-point 2-D sequence that is zero outside $0 \le x \le M-1$, $0 \le y \le N-1$, where M and N are integer powers of 2. In implementing the standard Walsh Transform of $f(x,y)$, we relate $f(x,y)$ to a new $M \times N$-point sequence $W(u,v)$.

iv) Define the sequence $W(u,v)$ in terms of $f(x,y)$.

The Walsh transform is defined as follows for two dimensional signals.

$$W(u,v) = \frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)\left[\prod_{i=0}^{n-1}(-1)^{(b_i(x)b_{n-1-i}(u)+b_i(y)b_{n-1-i}(v))}\right] \text{ or}$$

$$W(u,v) = \frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)(-1)^{\sum_{i=0}^{n-1}(b_i(x)b_{n-1-i}(u)+b_i(y)b_{n-1-i}(v))}$$

v) Define the concept of sequency in Walsh transform.

The concept of frequency exists also in Walsh transform basis functions. We can think of frequency as the number of zero crossings or the number of transitions in a basis vector and we call this number **sequency**. The Walsh transform exhibits the property of energy compaction as all the transforms that we are currently studying.

vi) Discuss at least three advantages of the Walsh transform.

1. Unlike the Fourier transform, which is based on trigonometric terms, the Walsh transform consists of a series expansion of basis functions whose values are only -1 or 1 and they have the form of square waves. These functions can be

implemented more efficiently in a digital environment than the exponential basis functions of the Fourier transform.

2. The forward and inverse Walsh kernels are identical except for a constant multiplicative factor of $\frac{1}{N}$ for 1-D signals.
3. The forward and inverse Walsh kernels are identical for 2-D signals. This is because the array formed by the kernels is a symmetric matrix having orthogonal rows and columns, so its inverse array is the same as the array itself.
4. For the fast computation of the Walsh transform there exists an algorithm called **Fast Walsh Transform (FWT).** This is a straightforward modification of the FFT. Advise any introductory book for your own interest.

c) Consider the population of random vectors $\underline{f}$ of the form

$$\underline{f} = \begin{bmatrix} f_1(x,y) \\ f_2(x,y) \\ \vdots \\ f_n(x,y) \end{bmatrix}.$$

Each component $f_i(x,y)$ represents an image. The population arises from their formation across the entire collection of pixels. Suppose that $n \geq 2$ and n even. Consider now a population of random vectors of the form

$$\underline{g} = \begin{bmatrix} g_1(x,y) \\ g_2(x,y) \\ \vdots \\ g_n(x,y) \end{bmatrix}$$

where the vectors $\underline{g}$ are the Karhunen-Loeve (KL) transforms of the vectors $\underline{f}$.

i) Derive the relationship between $\underline{g}$ and $\underline{f}$. Explain clearly how we formulate the transformation matrix of the KL transform.

The mean vector of the given population is defined as

$$\underline{m}_{\underline{f}} = E\{\underline{f}\}$$

The operator E refers to the expected value of the population, calculated theoretically using the probability density functions (pdf) of the elements f_i.

The covariance matrix of the population is defined as

$$\underline{\underline{C}}_{\underline{f}} = E\{(\underline{f} - \underline{m}_{\underline{f}})(\underline{f} - \underline{m}_{\underline{f}})^T\}$$

The operator E is now calculated theoretically using the probability density functions (pdf) of the elements f_i and the joint probability density functions between the elements f_i and f_j. It is proven (book work) that

$$\underline{\underline{C}}_{\underline{f}} = E\{\underline{f}\,\underline{f}^T\} - \underline{m}_{\underline{f}}\,\underline{m}_{\underline{f}}^T$$

Let $\underline{\underline{A}}$ be a matrix whose rows are formed from the eigenvectors of $\underline{\underline{C}}_{\underline{f}}$, ordered so that the first row of $\underline{\underline{A}}$ is the eigenvector corresponding to the largest eigenvalue, and the last row the eigenvector corresponding to the smallest eigenvalue. Eigenvalues are denoted by λ_i.

Suppose that $\underline{\underline{A}}$ is a transformation matrix that maps the vectors $\underline{f}$ into vectors $\underline{g}$ by using the following transformation

$$\underline{g} = \underline{\underline{A}}(\underline{f} - \underline{m}_{\underline{f}})$$

The above transform is called the **Karhunen-Loeve** transform.

iv) Discuss the type of the covariance matrix of $\underline{g}$ and the physical meaning of its elements.
It is proven (book work) that

$$\underline{m}_{\underline{g}} = \underline{0}$$

$$\underline{C}_{\underline{g}} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

The elements of the above matrix are the variances of the resulting image set.
The off-diagonal elements of the covariance matrix of the population of vectors $\underline{g}$ are 0, and therefore, the elements of the $\underline{g}$ vectors are uncorrelated.

v) Suppose that some elements of the covariance matrix of $\underline{g}$ are very small. Discuss the effectiveness of the KL transform in that case.
That means that the corresponsing images have very low variance and therefore, are almost uniform (they contain only weak edges). These images can be ignored.

iv) Suppose that the elements of the covariance matrix of $\underline{g}$ are very similar. Discuss the effectiveness of the KL transform in that case.
In case they are not small, that means all images carry important information and should be kept. Therefore, the KL transform is not of any use.

v) Suppose that a credible job could be done of reconstructing approximations to the n original images by using only half principal component images associated with the largest eigenvalues. What would be the mean square error incurred in doing so? Express your answer as a percentage of both the maximum and the minimum possible error.

$$e_{ms} = \sum_{j=n/2}^{n} \lambda_j$$

Maximum error $e_{\max} = \sum_{j=2}^{n} \lambda_j$

Minimum error $e_{\min} = \lambda_n$

2. a) A 4×4 grey scale original image passes through three spatial linear shift-invariant filters, resulting in three filtered images

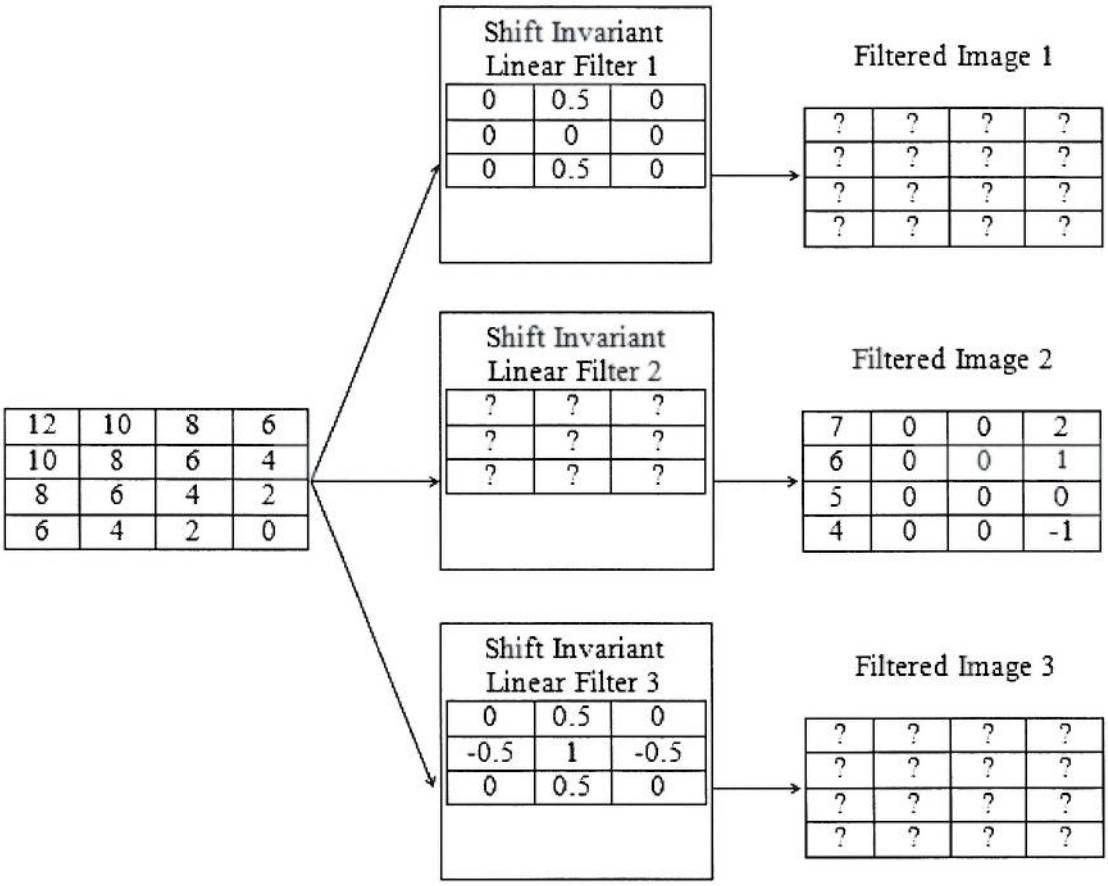


Figure 2.1

i) Compute Filtered Image 1 using zero-padding of the original image.

Filtered Image 1 is

Filtered Image 1

5	4	3	2
10	8	6	4
8	6	4	2
4	3	2	1

ii) Compute Filtered Image 3 using zero-padding of the original image.

Filtered Image 3 is

Filtered Image 3

12	4	3	4
16	8	6	5
13	6	4	2
8	3	2	0

iii) Based on relationship between Filtered Image 1, Filtered Image 2, and Filtered Image 3, determine the filter coefficients in Shift Invariant Linear Filter 2.

We see that Filtered Image 2= Filtered Image 3- Filtered Image 1, therefore,
Linear Filter 2=Linear Filter 3-Linear Filter 1

b) Give the linear filter masks for the following operations:

i) Region averaging.

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

ii) Weighted region averaging (with pixels near the centre given more weight).

$$\frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

iii) Emphasizing pixels with large differences from the background.
Laplacian or similar (book work).

iv) Detecting vertical (horizontal) edges.
Individual Sobel or related masks (book work).

v) Remove salt and pepper noise.
Median filter (book work).

vi) Image sharpening.
Any high pass filtering mask (book work).

vii) Image sharpening while preserving a proportion of the original.
High boost filtering (book work).

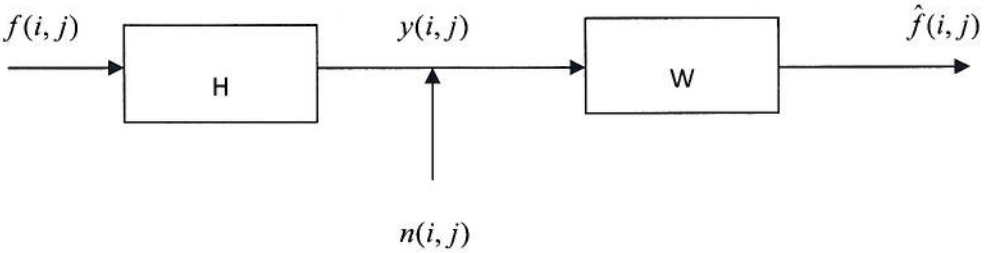
3. We are given the degraded version g of an image f such that in lexicographic ordering

$$g = Hf + n$$

where H is the degradation matrix which is assumed to be block-circulant, and n is the noise term which is assumed to be zero mean, independent and white. We propose to restore the degraded image using Wiener filtering.

a) i) Derive, the expressions for both the Wiener filter estimator and the restored image in both spatial and frequency domains and explain all symbols used.

The image restoration problem can be viewed as a system identification problem as follows:



The objective is to minimize the following function

$$E\{(\mathbf{f}-\hat{\mathbf{f}})^{\mathrm{T}}(\mathbf{f}-\hat{\mathbf{f}})\}$$

To do so the following conditions should hold:

1. $E\{\hat{\mathbf{f}}\} = E\{\mathbf{f}\} \Rightarrow E\{\mathbf{f}\} = \mathbf{W}E\{\mathbf{y}\}$
2. the error must be orthogonal to the observation about the mean

   $E\{(\hat{\mathbf{f}}-\mathbf{f})(\mathbf{y}-E\{\mathbf{y}\})^{\mathrm{T}}\} = 0$

From (i) and (ii) we have that

$$E\{(\mathbf{Wy}-\mathbf{f})(\mathbf{y}-E\{\mathbf{y}\})^{\mathrm{T}}\} = 0 \Rightarrow E\{(\mathbf{Wy}+E\{\mathbf{f}\}-\mathbf{W}E\{\mathbf{y}\}-\mathbf{f})(\mathbf{y}-E\{\mathbf{y}\})^{\mathrm{T}}\} = 0 \Rightarrow$$

$$E\{[\mathbf{W}(\mathbf{y}-E\{\mathbf{y}\})-(\mathbf{f}-E\{\mathbf{f}\})](\mathbf{y}-E\{\mathbf{y}\})^{\mathrm{T}}\} = 0$$

If $\tilde{\mathbf{y}} = \mathbf{y} - E\{\mathbf{y}\}$ and $\tilde{\mathbf{f}} = \mathbf{f} - E\{\mathbf{f}\}$ then

$$E\{(\mathbf{W}\tilde{\mathbf{y}}-\tilde{\mathbf{f}})\tilde{\mathbf{y}}^{\mathrm{T}}\} = 0 \Rightarrow E\{\mathbf{W}\tilde{\mathbf{y}}\tilde{\mathbf{y}}^{\mathrm{T}}\} = E\{\tilde{\mathbf{f}}\tilde{\mathbf{y}}^{\mathrm{T}}\} \Rightarrow \mathbf{W}E\{\tilde{\mathbf{y}}\tilde{\mathbf{y}}^{\mathrm{T}}\} = E\{\tilde{\mathbf{f}}\tilde{\mathbf{y}}^{\mathrm{T}}\} \Rightarrow \mathbf{W}\mathbf{R}_{\tilde{\mathbf{y}}\tilde{\mathbf{y}}} = \mathbf{R}_{\tilde{\mathbf{f}}\tilde{\mathbf{y}}}$$

If the original and the degraded image are both zero mean then

$\mathbf{R}_{\tilde{\mathbf{y}}\tilde{\mathbf{y}}} = \mathbf{R}_{\mathbf{yy}}$ and $\mathbf{R}_{\tilde{\mathbf{f}}\tilde{\mathbf{y}}} = \mathbf{R}_{\mathbf{fy}}$.

In that case we have that $\mathbf{W}\mathbf{R}_{\mathbf{yy}} = \mathbf{R}_{\mathbf{fy}}$.

If we go back to the degradation model and find the autocorrelation matrix of the degraded image then we get that

$$\mathbf{y} = \mathbf{Hf} + \mathbf{n} \Rightarrow \mathbf{y}^{\mathrm{T}} = \mathbf{f}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}} + \mathbf{n}^{\mathrm{T}}$$

$$E\{\mathbf{y}\mathbf{y}^{\mathrm{T}}\} = \mathbf{H}\mathbf{R}_{\mathbf{ff}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}_{\mathbf{nn}} = \mathbf{R}_{\mathbf{yy}}$$

$$E\{\mathbf{f}\mathbf{y}^{\mathrm{T}}\} = \mathbf{R}_{\mathbf{ff}}\mathbf{H}^{\mathrm{T}} = \mathbf{R}_{\mathbf{fy}}$$

From the above we get the following result

$$\mathbf{W} = \mathbf{R}_{\mathbf{fy}}\mathbf{R}_{\mathbf{yy}}^{-1} = \mathbf{R}_{\mathbf{ff}}\mathbf{H}^{\mathrm{T}}(\mathbf{H}\mathbf{R}_{\mathbf{ff}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}_{\mathbf{nn}})^{-1}$$

and the estimate for the original image is

$$\hat{\mathbf{f}} = \mathbf{R}_{\mathbf{ff}}\mathbf{H}^{\mathrm{T}}(\mathbf{H}\mathbf{R}_{\mathbf{ff}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}_{\mathbf{nn}})^{-1}\mathbf{y}$$

Note that knowledge of $\mathbf{R}_{\mathbf{ff}}$ and $\mathbf{R}_{\mathbf{nn}}$ is assumed.

In frequency domain

$$W(u,v)=\frac{S_{ff}(u,v)H^*(u,v)}{S_{ff}(u,v)\left|H(u,v)\right|^2+S_{nn}(u,v)}$$

$$\hat{F}(u,v)=\frac{S_{ff}(u,v)H^*(u,v)}{S_{ff}(u,v)\left|H(u,v)\right|^2+S_{nn}(u,v)}Y(u,v)$$

ii) The autocorrelation matrix of the original image f and of the additive noise n are necessary for Wiener filtering. Since both f and n are unknown, suggest a way to obtain estimates of them. Justify your answer.

The noise variance has to be known, otherwise it is estimated from a flat region of the observed image. In practical cases where a single copy of the degraded image is available, it is quite common to use $S_{yy}(u,v)$ as an estimate of $S_{ff}(u,v)$. **This is very often a poor estimate.**

b) Describe the Wiener smoothing filter.
In the absence of any blur, $H(u,v)=1$ and

$$W(u,v)=\frac{S_{ff}(u,v)}{S_{ff}(u,v)+S_{nn}(u,v)}=\frac{(SNR)}{(SNR)+1}$$

(a) $(SNR)>>1 \Rightarrow W(u,v)\cong 1$

(b) $(SNR)<<1 \Rightarrow W(u,v)\cong (SNR)$

(SNR) is high in low spatial frequencies and low in high spatial frequencies so $W(u,v)$ can be implemented with a lowpass (smoothing) filter.

c) Describe under what circumstances the Wiener filter is related to Inverse Filter.

If $S_{nn}(u,v)=0 \Rightarrow W(u,v)=\frac{1}{H(u,v)}$ which is the inverse filter

If $S_{nn}(u,v)\to 0$

$$\lim_{S_{nn}\to 0} W(u,v)=\begin{cases}\dfrac{1}{H(u,v)} & H(u,v)\neq 0\\ 0 & H(u,v)=0\end{cases}$$

which is the pseudoinverse filter.

4. a) Assume three grey scale images A, B, C which have 8 different grey levels denoted by $R_i, i = 0,\ldots,7$. The histograms of the three images are given in the table below.

Grey Level	Probability Image A	Probability Image B	Probability Image C
R_0	0.15	0.32	0.125
R_1	0.11	0.16	0.125
R_2	0.19	0.09	0.125
R_3	0.15	0.05	0.125
R_4	0.08	0.14	0.125
R_5	0.02	0.15	0.125
R_6	0.2	0.01	0.125
R_7	0.1	0.08	0.125

For the above images A, B, C, derive the Huffman code and answer the following questions:

i) What's the entropy of the original data source?

The entropy of each image is given by the relationship $H(S) = \sum_{i=0}^{7} p_i I(s_i) = -\sum_{i=0}^{7} p_i \log_2 p_i$.

For Image A this is 2.82756706 bits/symbol.

For Image B this is 2.64340315 bits/symbol.

For Image C this is 3 bits/symbol.

ii) What's the average coding length if using fixed-length coding?

3 bits per symbol.

iii) What's the average coding length of Huffman coding? (Need to show details of Huffman coding.)

For Image A this is 2.9 bits/symbol.

For Image B this is 2.72 bits/symbol.

For Image C this is 3 bits/symbol.

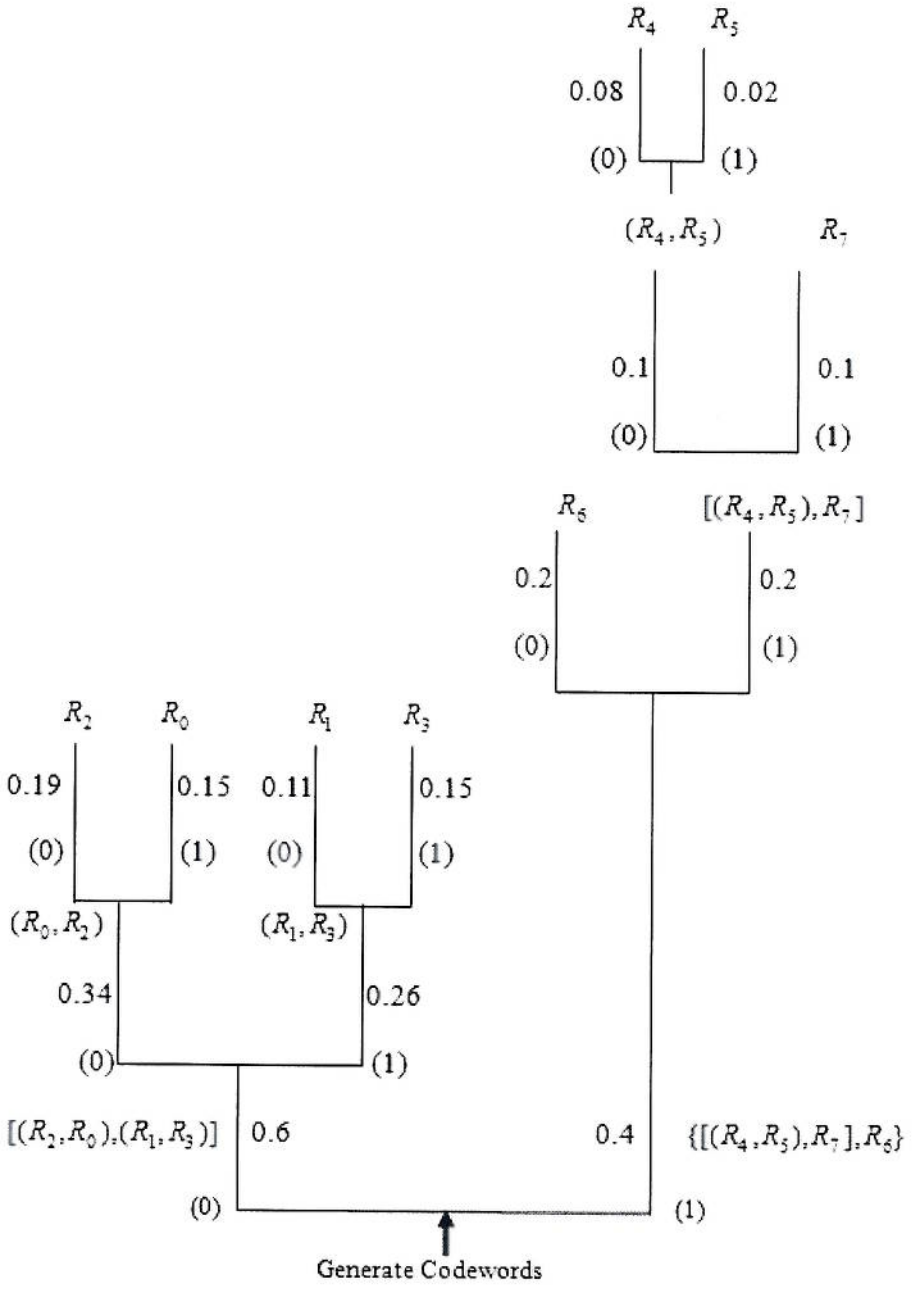
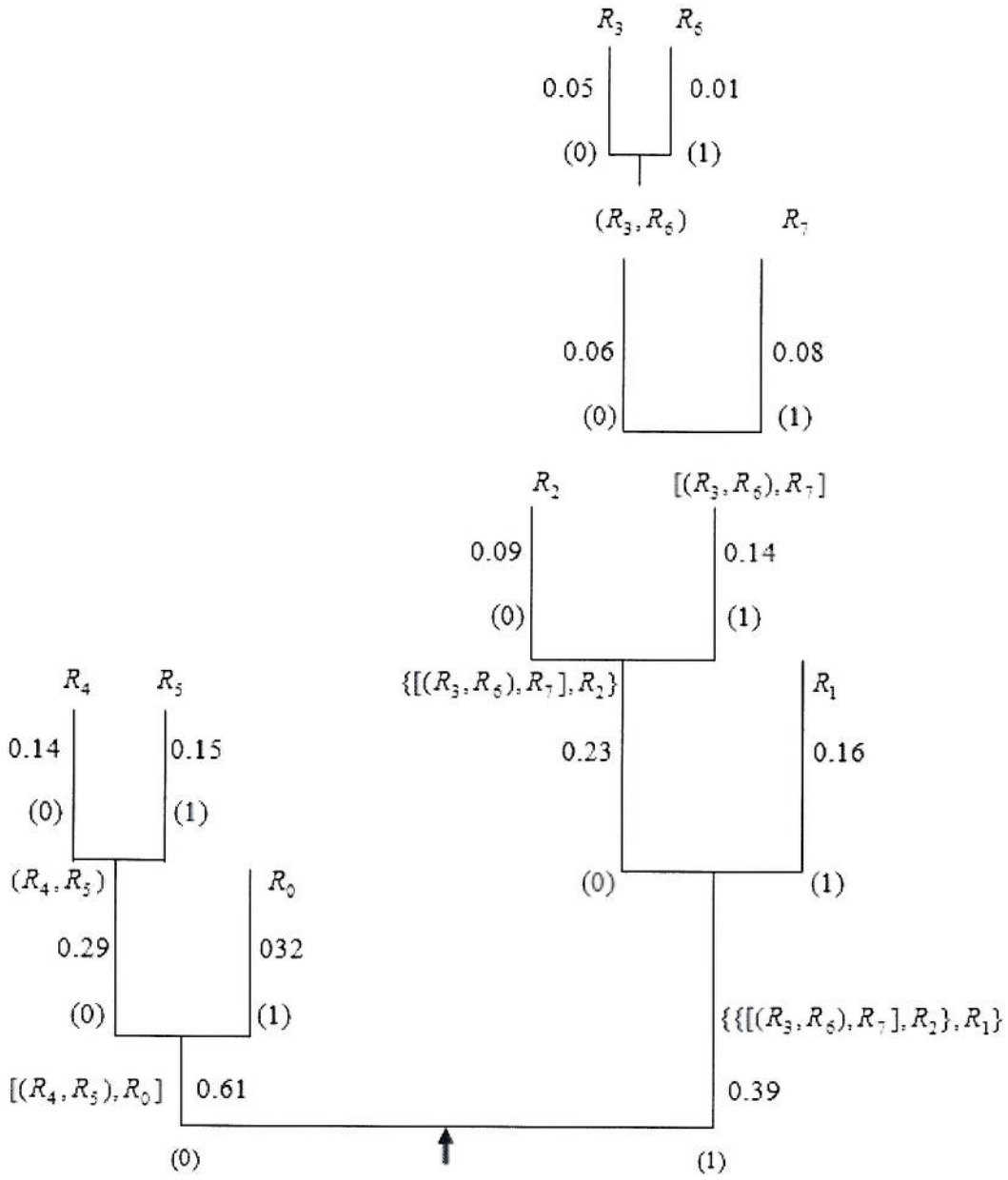


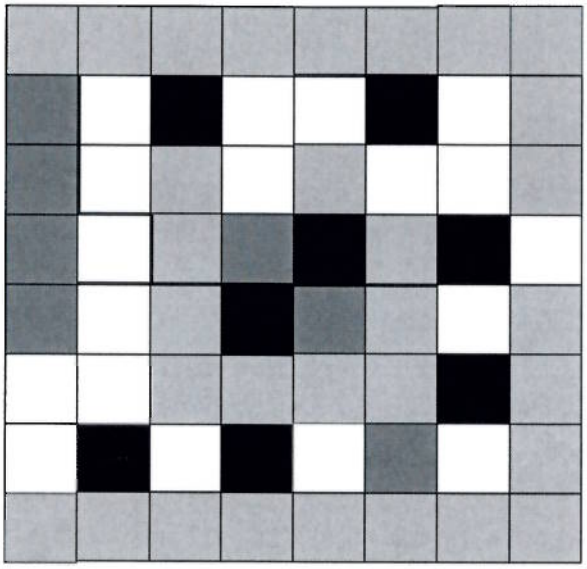
Figure: Huffman code of Image A



Generate Codewords

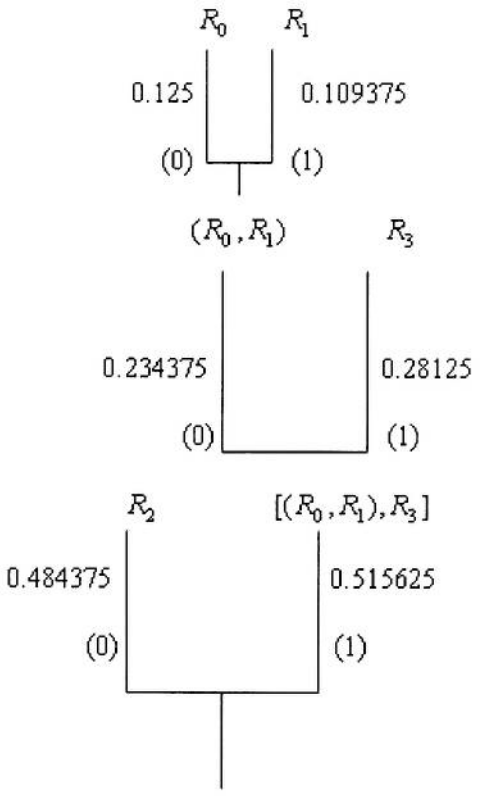
iv) Comment on relationship between probability combination, entropy and redundancy. Entropy in both cases is quite close to the fixed length number of bits per pixel. This is because there isn't any probability which is distinctively higher than the others. This phenomenon is more prominent in the first case. Based on the above comment the redundancies are quite small.

b) The following figure shows an 8x8 image with 4 different grey levels (black, dark grey, light grey, white).



i) Derive the probability of appearance (that forms the histogram) for each intensity level. Calculate the entropy of this image.
Total number of pixels: 64
Black pixels: 8
Dark grey pixels: 7
Light grey pixels: 31
White pixels: 18
We denote the grey levels as R_0=black, R_1=dark grey, R_2=light grey, R_3=white
$p_0 = 1/8$, $p_1 = 7/64$, $p_2 = 31/64$, $p_3 = 9/32$

ii) Derive the Huffman code.



iii) Calculate the average length of the fixed code and that of the derived Huffman code.

Fixed length: 2 bits/symbol
Huffman: 1.75 bits/symbol

iv) Calculate the ratio of image size (in bits) between using the fixed length coding and Huffman coding. Calculate the relative coding redundancy.
1.75/2
Redundancy=2-1.75=0.25